

LEUCHARS, United Kingdom

WMO: 031710

Lat: 56.377N

Lon: 2.863W

Elev: 38

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.2	27.8	20.0	15.0	29.9	22.5	16.9	32.3	40.2	45.6	35.3	45.8	5.5	270	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	71.6	61.5	68.8	59.9	66.5	58.5	63.4	68.9	61.7	66.3	60.3	64.3	13.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	80.5	65.6	59.7	76.6	63.8	58.4	72.9	62.3	28.8	68.8	27.6	66.5	26.6	64.4	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.1	27.1	23.8	20.2	77.4	4.4	3.5	17.0	79.9	14.4	81.9	12.0	83.9	8.8	86.5
			19.5	66.6	4.3	1.8	16.4	67.9	13.9	68.9	11.5	69.9	8.4	71.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	39.0	39.8	42.1	45.5	49.8	55.1	58.8	58.3	55.1	49.5	43.4	39.2	
	DBStd	8.58	4.93	4.98	5.04	4.71	4.71	3.98	3.75	3.57	3.94	4.83	5.31	6.11	
	HDD50	1707	341	288	248	149	62	5	0	0	5	68	206	335	
	HDD65	6202	805	706	709	584	471	296	197	209	299	479	649	798	
	CDD50	984	0	1	4	14	56	159	272	259	157	54	6	2	
	CDD65	7	0	0	0	0	0	0	4	2	1	0	0	0	
	CDH74	33	0	0	0	0	1	3	18	10	1	0	0	0	
	CDH80	2	0	0	0	0	0	0	1	1	0	0	0	0	
Wind	WSAvg	10.7	12.0	12.3	12.3	10.8	10.3	9.8	9.1	9.2	10.1	10.9	10.7	11.3	
Precipitation	PrecAvg	26.2	2.6	1.8	1.8	1.7	2.0	2.0	2.0	2.3	2.4	2.6	2.4	2.4	
	PrecMax	39.0	6.6	5.6	4.0	4.0	6.0	4.2	5.7	5.7	5.7	6.7	7.6	6.1	
	PrecMin	16.4	0.8	0.2	0.3	0.2	0.6	0.4	0.4	0.4	0.2	0.5	0.6	0.5	
	PrecStd	4.6	1.1	1.2	0.9	1.0	1.1	1.0	1.1	1.3	1.3	1.5	1.4	1.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	55.3	59.3	63.7	68.4	73.1	76.3	74.5	71.4	63.8	57.6	55.5	
		MCWB	50.6	51.3	51.3	54.0	58.2	60.9	64.5	63.3	62.1	57.4	54.0	52.6	
	2%	DB	51.4	52.6	55.3	59.3	63.8	68.6	72.0	70.1	67.2	60.8	55.3	52.9	
		MCWB	48.6	49.2	48.6	51.7	54.9	58.6	61.9	61.0	59.4	55.7	52.1	50.1	
	5%	DB	49.1	50.3	52.8	56.4	60.8	65.5	68.9	67.6	64.3	58.8	53.3	50.8	
		MCWB	46.2	46.9	47.3	49.4	53.0	56.8	60.3	59.5	57.2	54.7	50.2	48.3	
	10%	DB	47.0	48.1	50.5	53.9	58.2	63.0	66.3	65.3	62.1	56.8	51.5	48.4	
		MCWB	44.4	44.5	45.9	47.9	51.6	55.5	58.8	58.3	56.2	53.2	48.9	46.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	52.1	52.6	56.0	59.7	62.8	66.3	65.1	63.8	58.7	54.8	52.9
			MCDB	53.4	54.6	57.7	61.7	66.2	69.8	74.3	71.2	69.7	62.1	56.9	55.1
2%		WB	48.8	49.5	50.1	52.7	56.2	60.0	63.7	62.9	60.9	57.0	52.7	50.5	
		MCDB	51.2	52.2	54.3	57.9	61.8	65.9	69.5	67.7	64.9	59.8	54.7	52.6	
5%		WB	46.6	47.3	48.1	50.5	54.3	58.3	61.8	61.4	58.9	55.3	50.9	48.5	
		MCDB	48.8	50.0	51.8	55.2	59.2	63.5	66.7	65.3	62.5	57.8	52.7	50.4	
10%		WB	44.7	45.0	46.3	48.6	52.5	56.7	60.3	60.1	57.3	53.9	49.3	46.3	
		MCDB	46.7	47.5	50.0	52.8	56.7	61.4	64.5	63.4	60.6	56.4	51.2	48.2	
Mean Daily Temperature Range	5% DB	MDBR	9.2	10.7	12.3	13.1	13.6	13.1	13.7	13.8	13.0	11.8	10.6	9.5	
		MCDBR	11.1	12.0	15.2	17.0	17.5	17.7	18.0	16.8	16.2	13.0	11.6	11.4	
		MCWBR	9.9	9.7	10.3	10.5	10.1	9.8	9.6	9.2	9.8	9.3	9.9	10.3	
	5% WB	MCDBR	11.0	11.7	13.8	15.7	15.9	15.9	16.0	14.0	14.2	11.8	11.0	11.1	
		MCWBR	10.1	9.9	10.3	10.3	10.1	9.8	9.6	8.9	9.7	9.4	9.9	10.3	
Clear-Sky Solar Irradiance	taub	0.317	0.334	0.365	0.387	0.384	0.385	0.386	0.373	0.358	0.342	0.315	0.299		
	taud	2.272	2.366	2.313	2.282	2.339	2.364	2.368	2.414	2.465	2.443	2.338	2.234		
	Ebn at Noon	180	230	251	263	271	272	269	265	253	225	181	154		
	Edh at Noon	17	23	30	36	36	36	35	31	26	21	16	14		
All-Sky Solar Radiation	RadAvg	156	362	685	1128	1507	1484	1429	1139	800	429	204	115		
	RadStd	23	33	85	119	95	125	125	78	52	49	22	14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page